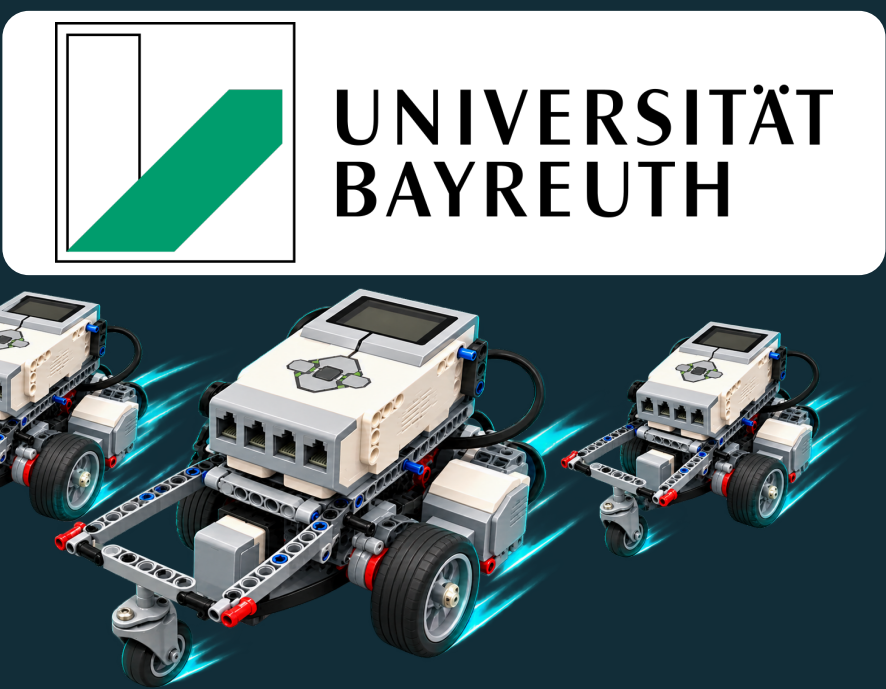


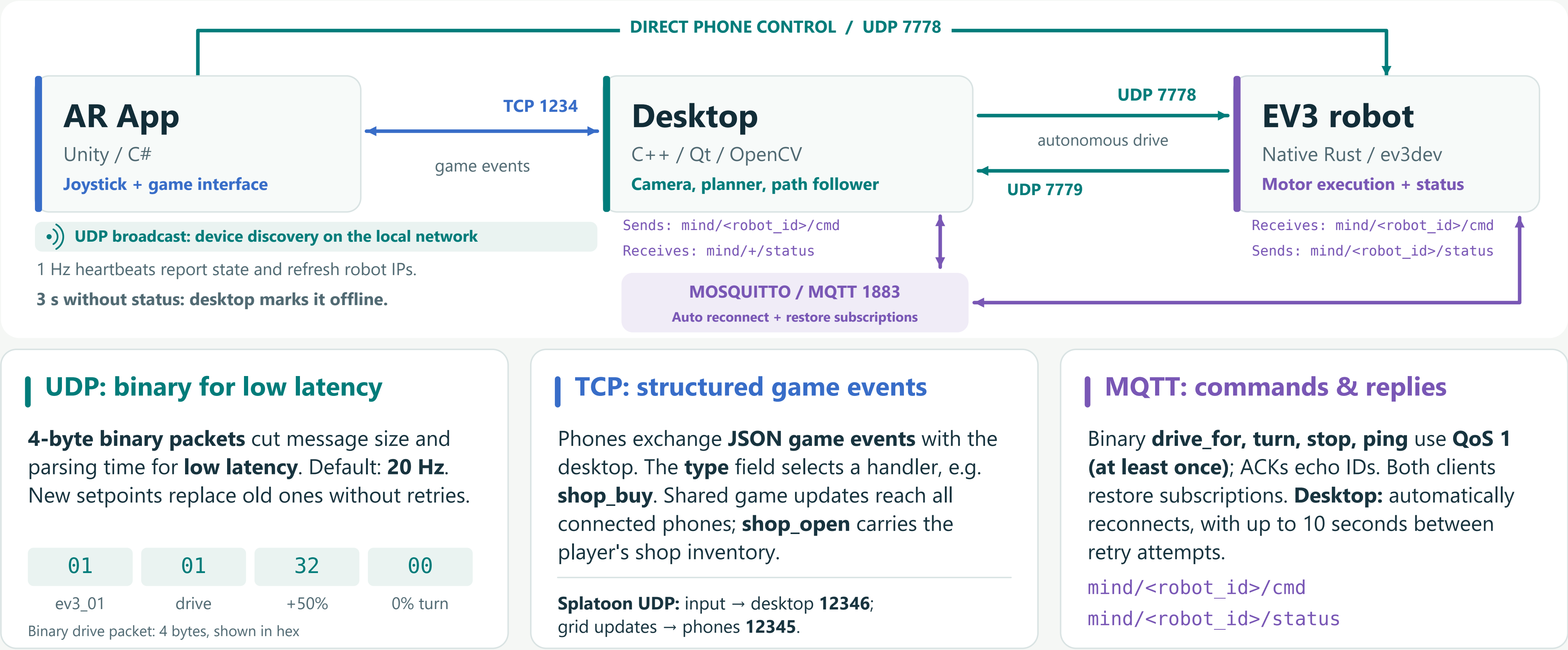


Robot Control & Communication

Collision avoidance, steering & network protocols

01 One system. Three communication channels.

ARCHITECTURE



02 From a target to controlled movement

DESKTOP CONTROL

Plan around other robots

Global board routes + local collision avoidance

ROUTE

YIELD

CLEAR A NODE

- 1 / Route.** Dijkstra selects the board-node route. Grid A* detours around parked robots; clearance-checked corner rounding smooths the waypoints.
- 2 / Yield.** A central coordinator watches nearby paths. The higher robot ID yields, replans around the other robot, or holds if no detour exists.
- 3 / Clear.** If a destination is occupied, the blocker is routed to a free parking spot off the lanes. The arriving robot waits for the node to clear.

Follow the path, correct the heading

Camera pose → lookahead target → steering → wheels

planned path

20 cm lookahead

heading error

- 1 / Observe.** ArUco markers provide position and heading. A board homography converts camera pixels into physical centimetres.
- 2 / Steer.** A lookahead point sets the heading error. A PID controller computes turn rate; forward speed falls in corners and near the goal.
- 3 / Settle.** Sharp turns use pivot pulses and pauses for camera feedback to catch up. Calibration scales compensate drive and turn bias.



03 Inside the brick

RUST + EV3DEV

Ferris at the wheel

Four workers, one motor controller

Native Rust runs directly on the EV3's Linux system. A mutex protects shared drive state; each valid UDP packet overwrites the setpoint and wakes the motor worker. A 10 ms timeout keeps watchdog checks running between packets.

UDP
Decode + ID filter
Wake the motor loop

MOTOR
10 ms check interval
Regulated B/C motors

MQTT
Dispatch commands
Interruptible moves

HEARTBEAT
1 Hz to desktop
State + battery + ID

Two ways to move: stream speed + turn or independent wheel speeds; queue **drive_for** and **turn** for distance and angle moves. The motor worker sends the completion ACK.

Differential drive: $v_L = v - \omega b/2$; $v_R = v + \omega b/2$.
 v : forward speed; ω : turn rate; b : wheel spacing.
At the speed limit, both wheels slow together to keep the curve.
Motors B/C; 56 mm wheels; $b = 126$ mm.

Independent stop mechanisms

150 ms No fresh drive packet: the EV3 watchdog stops continuous motion.

400 ms Stale camera pose: the desktop commands zero drive.

STOP MQTT sets an interrupt flag for precision moves; motors poll it.